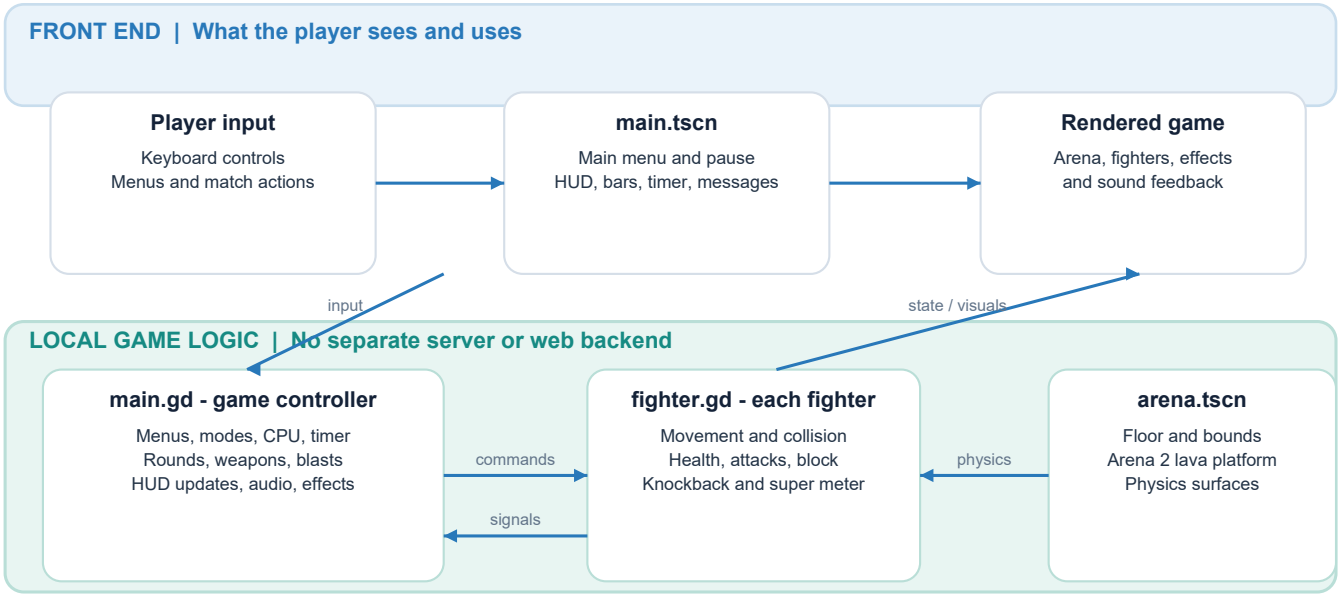
STICK ARENA

Front-end and game-logic diagram + test cases and runtime results

40 passed	0 failed	Godot 4.7.2

System architecture

This is a local Godot game. The term backend below means the game logic running inside the game; there is no separate web server, database, or online service.



How the pieces connect

The player uses keyboard input through the game scene. main.gd coordinates menus, modes, CPU behavior, rounds, timer, weapons, projectiles, HUD, audio, and effects. It sends actions to each fighter.gd; fighters report health, hits, knockouts, meter changes, and weapon drops back with signals. The arena scene supplies the floor, stage geometry, and collision surfaces.

Project files: scenes/game/main.tscn, scenes/arenas/arena.tscn, scenes/player/fighter.tscn, scripts/game/main.gd, scripts/player/fighter.gd.

Runtime test summary

The automated headless smoke harness directly exercises game methods and menu handlers. It is useful for core logic regression, but it does not fully replace playing the game with a keyboard or checking rendered visuals and sound.

Check	Result
Automated smoke harness	40 passed, 0 failed
Main-scene launch	90 frames; process exited with code 0; no scene/script errors
Environment warning	Godot could not read the Windows root-certificate store. This did not stop either run.
Manual verification remaining	Physical keyboard controls, movement/collision play, settings, full timer presentation, visual effects, and audible sound still need live review.

Result meanings

Pass means the documented behavior was exercised in the smoke harness. Partial pass means only the noted portion was checked; remaining steps need a live/manual test. Not run means the behavior was not exercised by this runtime run.

How to rerun

Open a terminal in the project folder and run the Godot 4.7.2 executable in headless mode with the project path and --script res://tests/runtime_smoke.gd. For a launch-only smoke check, use --headless --path <project-folder> --quit-after 90.

Test cases and results

Run notes are from the 2026-10-09 headless test. For every partial or not-run case, use the procedure below in the game to complete the check. Record a date and any observed issue when doing manual testing.

Menu and setup

ID	Test case / procedure / expected result	Result
MENU-01	Main menu navigation Steps: Launch the game. Open Play, Settings, and Controls with Up/Down and Enter. Use Esc to return from submenus. Expected: The main menu appears first; menu navigation and back behavior work. Run note: Launch menu and Play handler tested; physical navigation and Esc not tested.	Partial pass
MENU-02	Mode and stage selection Steps: Choose CPU difficulty, 2 Players, or Practice. Select an opponent where offered, then choose Arena 1 or Arena 2. Expected: All intended modes and stages lead to the correct match setup. Run note: Practice-to-dummy-to-stage flow and stage toggle tested; keyboard navigation and CPU choices not tested.	Partial pass
MENU-03	Settings and controls reference Steps: Change master volume and screen-shake toggle. Open Controls and compare the displayed keys. Expected: Volume affects sound; shake toggle works; controls reference is accurate. Run note: Settings/audio behavior and navigation need interactive verification.	Not run

Normal match

ID	Test case / procedure / expected result	Result
MATCH-01	Movement and Arena 1 bounds Steps: In a two-player match, move both fighters to arena edges and toward each other; jump and land. Expected: Correct keys move each fighter; fighters face each other, do not pass through, stay in bounds, and land on the floor. Run note: Movement, physical collisions, and edge behavior not covered by the smoke harness.	Not run
MATCH-02	Punch, damage number, cooldown Steps: Land an unblocked punch, then rapidly tap or hold attack. Expected: Punch removes 20 health, shows applied damage, and respects the existing cooldown. Run note: 20 damage, damage label, and cooldown tested; reach/facing not tested.	Partial pass
MATCH-03	Block and chip damage Steps: Block Player 1 with S and Player 2 with Down; receive a normal punch. Expected: Blocked punch removes 5 health, shows -5, and causes no knockback. Run note: 5 chip damage and no knockback tested.	Pass
MATCH-04	Bat pickup, attack, and disarm Steps: Walk over the bat, attack while holding it, then take a hit while armed. Expected: Pickup appears in hand and on HUD; bat hit removes 35; a hit disarms the holder. Run note: Equip state, 35 damage, and disarm/drop tested; pickup overlap and recollection not tested.	Partial pass
MATCH-05	Super meter and projectile Steps: Build meter, fire a special, test blocked/unblocked damage, and attempt a second blast while one is active. Expected: Meter fills and resets; damage is 40/10; only one blast per fighter; effects and sounds appear. Run note: Damage, firing, meter consumption, trail node, and one-blast rule tested; physical input, impact, and sound/visual quality not tested.	Partial pass
MATCH-06	Round timer and timeout Steps: Let the round timer reach zero with equal and unequal health; pause during a live round. Expected: Timer starts with FIGHT, stops when appropriate, displays TIME!, resolves higher health, and replays a draw. Run note: Timeout, draw, and higher-health winner logic tested; full countdown, red pulse, and TIME display timing not tested.	Partial pass
MATCH-07	Best-of-three match end Steps: Win two rounds, then repeat with the other player as winner. Expected: Win indicators update and the first fighter to two round wins gets the match-end screen. Run note: Two round wins reaching match end tested.	Pass
MATCH-08	Pause, resume, restart, main menu Steps: Press Esc during a match and try Resume, Restart, and Main Menu. Expected: Pause options work; timer and gameplay stop while paused. Run note: Pause/resume states tested; key navigation, restart, and main-menu action not tested.	Partial pass

Stages and CPU

ID	Test case / procedure / expected result	Result
STAGE-01	Arena 1 floor and no hazard Steps: Play Arena 1 and move or get knocked near either edge. Expected: Full-width floor and current boundary behavior remain; no lava damage occurs. Run note: Arena 1 visual and physical boundary behavior not exercised.	Not run
STAGE-02	Arena 2 lava and weapon respawn Steps: Walk or get knocked into lava; drop the bat into a pit and wait for respawn. Expected: Lava causes periodic damage and recoil; bat disappears and returns to safe platform after delay. Run note: Lava damage, stage physics toggle, and weapon loss scheduling tested; actual fall, intervals, effects, and respawn completion not tested.	Partial pass
CPU-01	CPU behavior and difficulty Steps: Play Easy, Normal, and Hard; test weapon seeking and CPU special use at range. Expected: CPU difficulty changes reaction/block/attack behavior; CPU uses weapons and full-meter special appropriately. Run note: Arena 2 edge-avoidance logic tested; randomized combat, difficulty, weapon seeking, and special timing not tested.	Partial pass

Practice mode and presentation

ID	Test case / procedure / expected result	Result
PRACTICE-01	Practice mode selection Steps: Choose Practice, select CPU or dummy and difficulty, then select each stage. Expected: Practice offers opponent choices and both arenas start correctly. Run note: Practice-to-dummy-to-stage handler path tested; CPU difficulty selection not tested.	Partial pass
PRACTICE-02	No timer or round win Steps: Start Practice and continue through a KO. Expected: No countdown or round/match win; the bout continues. Run note: Hidden/stopped timer and no awarded round on KO tested.	Pass
PRACTICE-03	Passive dummy behavior Steps: Wait while facing the dummy, then hit and knock it back. Expected: Dummy does not move or attack by itself but can be hit and moved. Run note: Dummy move direction and attack disabled state tested.	Pass
PRACTICE-04	Manual super-meter fill Steps: Press M in Practice; verify normal matches do not allow manual fill. Test N only if P2 is human-controlled. Expected: Practice key fills the intended fighter meter only in Practice. Run note: Physical M key not verified by headless input. Current Practice offers CPU/dummy, so human P2/N path is unavailable.	Not run
PRACTICE-05	Recovery after lethal damage Steps: Reduce a fighter to zero health and time recovery; repeat for both fighters. Expected: KO fighter restores to full health after about one second without ending Practice. Run note: One-second recovery scheduling and full-health restore tested.	Pass
PRACTICE-06	Applied damage numbers Steps: Test punch, block, bat, special, and lava hits, including low remaining health. Expected: Damage number reflects the actual health removed for each hit. Run note: Punch and lava labels tested; other label values not tested.	Partial pass
PRACTICE-07	Pause and restart in Practice Steps: Pause, wait, resume, and test Restart and Main Menu. Expected: Pause flow works; timers/recovery stop while paused; restart/menu return correctly. Run note: Interactive pause/restart/menu behavior not tested.	Not run
VISUAL-01	Visual and audio feedback Steps: Review both arenas, hits, blocks, specials, KO, settings, and sound effects in a live match. Expected: Bars, indicators, hit feedback, particles, stages, camera, and sounds look and respond correctly. Run note: Visual quality and audible output need an in-game review.	Not run

Controls reference

Context	Controls
Menus	Up/Down navigate; Enter confirms; Left/Right changes settings or stage; Esc goes back or pauses; R restarts.
Player 1	A/D move; W or Space jump; F attack; S block; G special.
Player 2	Left/Right move; Up jump; O attack; Down block; P special.
Practice meter	M fills Player 1 meter. N is described for Player 2 when human-controlled; the current Practice opponent choices are CPU or dummy, so that route is not currently selectable.

Manual result recording

For hands-on checks, record Pass, Fail, or Blocked for each case and add a short note (for example, exact key sequence, observed damage, screenshot location, or error message).